

Carbonate Diagenesis, Cement Fabrics & Syntaxial Overgrowths

Comprehensive Textbook Study Guide & Complete Solution for DGMS 2019 Exam (Question 2)

Standard Reference: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Edition, 2006)

PART I

Theoretical Foundation & Direct Textbook Analysis

Carbonate diagenetic environments, cement morphologies, optical continuity, and crystallographic overgrowth mechanisms

1. CARBONATE DIAGENETIC ENVIRONMENTS & CEMENT PRECIPITATION

Carbonate sediments are extraordinarily susceptible to post-depositional physical and chemical alteration (diagenesis) because calcium carbonate minerals (aragonite, high-Mg calcite, low-Mg calcite) are chemically reactive and metastable in natural waters (Boggs, 2006, Chapter 6).

"Carbonate diagenesis takes place in three major realms: (1) the marine environment, (2) the meteoric environment (subdivided into the vadose unsaturated zone and the phreatic saturated zone), and (3) the deep-burial environment. Cementation is the primary process responsible for the lithification of loose carbonate grains into rigid limestone. The mineralogy and morphology of carbonate cements are diagnostic of the chemistry, saturation state, and hydrologic flow of pore waters."

— Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), pp. 204–207

The morphology of carbonate cement crystals ranges widely:

- **Acicular / Aragonite Needles & Fibrous Cements:** Common in the marine phreatic zone (high Mg/Ca ratio favors aragonite and high-Mg calcite).
- **Meniscus & Pendant (Gravitational) Cements:** Confined to the meteoric vadose zone, where capillary forces draw pore water into contact points between grains (meniscus) or beneath grains (stalactitic droplets).
- **Equant / Blocky Calcite Cements:** Interlocking crystals of equal dimensions, characteristic of freshwater meteoric phreatic and deep burial realms.
- **Syntaxial Rim Overgrowths:** Single-crystal cement jackets precipitated in crystallographic and optical continuity around monocrystalline host grains.

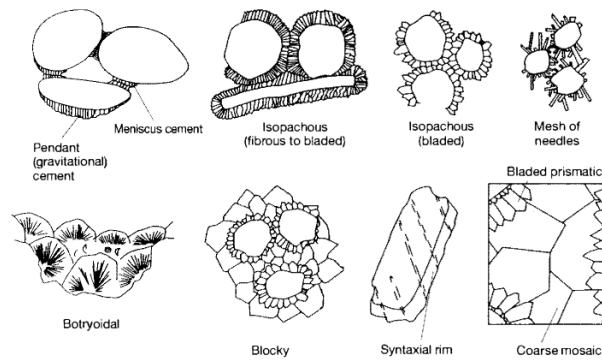


Figure 6.15

Principal kinds of cements that form in carbonate rocks during diagenesis. **Seafloor diagenetic environments** are characterized particularly by aragonitic meniscus and pendant cements (in beachrock), isopachous cement, needle cement, and botryoid cement. **Meteoric-realm** cements are composed dominantly of calcite and include rhus and pendant cements in the **vadose zone** and isopachous, blocky, and syntaxial cements in the **phreatic zone**. Cements of the **subsurface burial realm** are also calcite and include syntaxial rims, bladed prismatic, and coarse mosaic types. [Mc from James, N. P., and P. W. Choquette, 1983, *Geoscience Canada*, v. 10, Fig. 3, p. 1984, *Geoscience Canada*, v. 11, Fig. 24, p. 177; 1987, *Geoscience Canada*, v. 14, p. 21, p. 16.]

In the meteoric realm, dissolution is a more important process than cementation; however, cementation does occur. The cement is almost exclusively calcite. As mentioned, the calcium carbonate that forms this cement is derived by dissolution of less stable aragonite and high-magnesian calcite. In the (water-) unsaturated vadose zone, calcite cements are commonly meniscus and pendant cements. In the water-saturated phreatic zone, they are isopachous, blocky, or syntaxial.

Figure 6.15 (Boggs, *Principles of Sedimentology and Stratigraphy*, 2006, p. 208): Morphology and distribution of carbonate cement types. Note the distinctive **Syntaxial rim cement** surrounding a monocrystalline echinoderm grain, alongside meniscus, pendant, isopachous, and blocky mosaic fabrics.

2. THE MECHANISM OF SYNTAXIAL OVERGROWTH & OPTICAL CONTINUITY

The word *syntaxial* derives from the Greek *syn* (together) and *taxis* (order/arrangement). In sedimentary petrography, it defines cement growth that shares the exact crystal lattice orientation of the substrate.

"In the water-saturated phreatic zone, cements are isopachous, blocky, or syntaxial rim cements. Syntaxial rims form by precipitation of optically continuous calcite cement on single-crystal echinoderm fragments. Because syntaxial overgrowths are optically continuous with the original crystal, the host grain and the cement extinguish simultaneously under crossed polars on the petrographic microscope."

— Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy* (4th Ed., 2006), pp. 141, 208

Echinoderm plates (crinoid ossicles, echinoid spines) are unique in paleontology because each individual skeletal element behaves as a **single giant crystal of high-Mg calcite**.

When pore waters become supersaturated with respect to low-Mg calcite:

- The echinoderm surface acts as an ideal, energetically favored seed template (epitactic nucleation site).
- Calcite precipitates without needing to nucleate new crystallographic centers, growing outward as an optical continuation of the host lattice.

- Under a petrographic microscope between crossed polars, when the stage is rotated, **both the core echinoderm fragment and the surrounding syntaxial cement extinguish at the exact same angle.**
- The boundary between the original grain and the syntaxial overgrowth is typically demarcated only by a thin "dust line" of micro-inclusions or iron oxide coating.

3. COMPARATIVE TERMINOLOGY: SYNTAXIAL, ANTITAXIAL, BLOCKY & RIM

Competitive examinations frequently test precise terminology differentiating overgrowth types:

- **Syntaxial Overgrowth:** Mineral growth where cement continues the lattice and optical orientation of the substrate (crystal axis [c-axis] is identical). Diagnostic of echinoderm-bearing limestones (encrinites) and syntaxial quartz overgrowths on detrital quartz grains in quartz arenites.
- **Antitaxial Growth:** A term strictly from *structural geology and vein mechanics* (Durney & Ramsay, 1973). Antitaxial veins are fibrous veins where fibrous mineral crystals grow outward from a central median suture line towards the vein walls (crystals grow at the vein wall-rock contact). It does *not* describe carbonate grain cementation.
- **Blocky / Equant Cement:** A mosaic of randomly oriented calcite crystals that have approximately equal dimensions and vary randomly in their optical extinction angles.
- **Rim Cement (Generic):** An isopachous or microcrystalline crust coating a grain. While syntaxial overgrowth forms a "rim", calling it merely "rim overgrowth" fails to specify the critical defining condition: *optical continuity with the single crystal substrate.*

PART II

DGMS 2019 Examination: Question & Worked Solution

Rigorous option-by-option critical analysis and definitive textbook-grounded answer

ORIGINAL QUESTION — DGMS 2019 (JUNIOR SCIENTIFIC OFFICER)

Q.2. In carbonate diagenesis, if calcite cement is overgrown on a single calcite crystal in same optical orientation, it is referred to as

- (a) Rim overgrowth
- (b) Antitaxial overgrowth
- (c) Blocky overgrowth
- (d) Syntaxial overgrowth

Correct Answer: (d) Syntaxial overgrowth



When cement precipitates around a monocrystalline grain (such as an echinoderm fragment or quartz grain) in identical crystallographic and optical orientation, it is defined scientifically as a syntaxial overgrowth (or syntaxial rim cement).

OPTION-BY-OPTION CRITICAL ANALYSIS

✗ (a) Rim overgrowth

INCORRECT

Why it is incorrect: "Rim overgrowth" or "isopachous rim" is a generic descriptive term denoting any cement layer that coats a grain boundary. A rim can be composed of randomly oriented microcrystalline calcite, isopachous fibrous aragonite needles, or tangential coatings, none of which share the optical orientation of the host grain. It misses the specific crystallographic relationship defined in the question.

✗ (b) Antitaxial overgrowth

INCORRECT

Why it is incorrect: "Antitaxial" is a structural geology term describing fibrous vein-fill dynamics where fibers grow outward from a median line towards the wall rock (Durney & Ramsay, 1973). In diagenetic sedimentology, cement growing outward from a detrital grain into pore space is *syntaxial*, never antitaxial.

✗ (c) Blocky overgrowth

INCORRECT

Why it is incorrect: Blocky cement (equant spar) consists of equant, polygonal calcite crystals that nucleate randomly in pore spaces without crystallographic alignment to the host grain. Each crystal in a blocky mosaic has its own independent optical orientation.

✓ (d) Syntaxial overgrowth

CORRECT

Why it is correct: As defined verbatim in Boggs (2006, pp. 141, 208), precipitation of mineral cement on a single-crystal substrate (e.g., echinoderm ossicle or quartz grain) where the new cement adopts the **exact same optical and crystallographic orientation** as the host crystal is termed **syntaxial overgrowth**. Both host and cement extinguish simultaneously under crossed polars.

PART
III

Synthesis Reference Tables & High-Yield Practice Questions

Quick-reference diagnostic tables and 8 curated exam questions for DGMS, UPSC GSI, and IIT-JAM

SYNTHESIS TABLES: CARBONATE CEMENTS & OVERGROWTH FABRICS

Table 1: Carbonate Cement Fabrics and Diagenetic Environments

Cement Morphology	Mineralogy	Diagenetic Environment	Diagnostic Feature
Syntaxial Rim Overgrowth	Low-Mg Calcite	Meteoric Phreatic & Burial	Optical continuity with single-crystal host (echinoderm)
Isopachous Fibrous Rim	Aragonite / High-Mg Calcite	Marine Phreatic	Uniform isopachous coating of needles perpendicular to grain
Meniscus Cement	Low-Mg Calcite	Meteoric Vadose	Blunt, curved meniscus bridges at grain contacts
Pendant (Microcodium/Gravitational)	Low-Mg Calcite	Meteoric Vadose	Thickens on undersides of grains due to hanging droplets
Blocky / Equant Mosaic	Low-Mg Calcite	Meteoric Phreatic & Burial	Equant polygonal crystals increasing in size towards pore center
Poikilotopic Spar	Calcite / Ferroan Calcite	Deep Burial	Very large calcite crystals enclosing multiple detrital grains

Table 2: Comparison of Growth Terms in Geology

Term	Discipline	Growth Substrate & Direction	Optical Property
Syntaxial	Sedimentology / Petrology	Grows outward from host grain into open pore space	Identical optical extinction to host grain
Antitaxial	Structural Geology	Vein fibers grow outward from median line to wall rock	Independent fiber optical axes
Epitaxial	Mineralogy / Materials	Overgrowth of mineral A onto mineral B with lattice match	Distinct optical properties from substrate

Q1. Syntaxial calcite overgrowths in limestones are most commonly developed around:

- (a) Ooids (b) Echinoderm fragments (c) Peloids (d) Intraclasts

→ **Correct Answer: (b) Echinoderm fragments**

Explanation: Each echinoderm skeletal element (crinoid plate, echinoid spine) is a single, optically uniform crystal of calcite, serving as an ideal template for syntaxial rim precipitation.

Q2. Under crossed polars, a syntaxial overgrowth and its host grain:

- (a) Show 90° difference in extinction angles
(b) Extinguish simultaneously at the exact same orientation
(c) Display undulose versus uniform extinction
(d) Never extinguish

→ **Correct Answer: (b) Extinguish simultaneously at the exact same orientation**

Explanation: Optical continuity means the crystallographic c-axes of both host and cement are parallel, resulting in simultaneous extinction.

Q3. Meniscus and pendant cement fabrics in carbonate rocks are diagnostic indicators of:

- (a) Deep marine phreatic environment
(b) Meteoric vadose environment
(c) Hydrothermal burial realm
(d) Anoxic deep marine basin

→ **Correct Answer: (b) Meteoric vadose environment**

Explanation: Vadose environments contain both air and water in pores; surface tension holds water in menisci at grain contacts or hanging beneath grains, depositing meniscus and pendant cements.

Q4. Syntaxial overgrowths in sandstones are most prominently exhibited by which mineral?

- (a) Plagioclase (b) Quartz (c) Biotite (d) Tourmaline

→ **Correct Answer: (b) Quartz**

Explanation: In quartz arenites, silica cement precipitates around detrital quartz grains in optical continuity, forming syntaxial quartz overgrowths with euhedral crystal faces.

Q5. The "dust line" frequently visible between a host grain and its syntaxial rim represents:

- (a) A zone of high strain
(b) Impurities, iron oxides, or micro-inclusions on the original grain surface
(c) Chemical replacement by dolomite
(d) Fracturing during compaction

→ **Correct Answer: (b)**

Explanation: Prior to cementation, the detrital grain surface accumulates micro-coatings of clay, iron oxides, or fluid inclusions, which remain preserved as a dark line separating the host from the new cement.

Q6. Isopachous fibrous aragonite needle cement is typical of:

- (a) Subaerial caliche crusts
- (b) Warm, shallow marine phreatic environments
- (c) Deep burial diagenesis
- (d) Meteoric phreatic lenses

→ **Correct Answer: (b) Warm, shallow marine phreatic environments**

Explanation: Marine waters have high Mg/Ca ratios that inhibit calcite growth and promote precipitation of needle-like aragonite or high-Mg calcite as uniform rims around grains.

Q7. In structural geology, fibrous veins that grow from a central median suture outwards towards the wall rocks are termed:

- (a) Syntaxial
- (b) Antitaxial
- (c) Composite
- (d) Ataxial

→ **Correct Answer: (b) Antitaxial**

Explanation: Antitaxial veins grow at the vein/wall-rock interface with the oldest crystal material in the center, contrasting with syntaxial veins where crystals grow inward toward the vein center.

Q8. A limestone dominated by crinoid fragments cemented by syntaxial rim spar is termed:

- (a) Micrite
- (b) Encrinite
- (c) Oolite
- (d) Travertine

→ **Correct Answer: (b) Encrinite**

Explanation: Encrinites are crinoidal packstones or grainstones where vast volumes of syntaxial calcite cement interlock around abundant crinoidal debris.